

File permissions in Linux

Project description

Linux commands help manage and sort through files and directories. Permissions help establish who can access the files and directories and how much control they have over them.

Check file and directory details

To check the file permissions for the projects directory, I first changed directory into projects. I then used `ls -l` to list the permissions within the projects directory, as shown below.

```
researcher2@b37263ddfdd4:~$ ls
projects
researcher2@b37263ddfdd4:~$ cd projects
researcher2@b37263ddfdd4:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Oct  7 09:43 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Oct  7 09:43 project_k
.txt
-rw-r----- 1 researcher2 research_team  46 Oct  7 09:43 project_m
.txt
-rw-rw-r-- 1 researcher2 research_team  46 Oct  7 09:43 project_r
.txt
-rw-rw-r-- 1 researcher2 research_team  46 Oct  7 09:43 project_t
.txt
researcher2@b37263ddfdd4:~/projects$
```

Describe the permissions string

For the `project_k.txt` file, the permissions seen are `-rw-rw-rw-`. To break this down, the 10 character string can be split into three sections for read, write and execute permissions for the user, group and other within the file. `Project_k.txt` includes read and write permissions for the user, group and other and includes no execute permissions.

Change file permissions

The project_k.txt file should not have any write permissions for others. As a result, I have used `chmod o-w project_k.txt` to change the other permissions for project_k.txt. This command shows that the o, which stands for others, has had the write permissions removed with the `-w` command next to it.

```
researcher2@b37263ddfdd4:~/projects$ chmod o-w project_k.txt
researcher2@b37263ddfdd4:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Oct  7 09:43 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Oct  7 09:43 project_k
.txt
-rw-r----- 1 researcher2 research_team  46 Oct  7 09:43 project_m
.txt
-rw-rw-r-- 1 researcher2 research_team  46 Oct  7 09:43 project_r
.txt
-rw-rw-r-- 1 researcher2 research_team  46 Oct  7 09:43 project_t
.txt
researcher2@b37263ddfdd4:~/projects$
```

Change file permissions on a hidden file

The .project_x.txt file has been archived and as such is a hidden file. It should not have any write permissions at all, but the user and group should have read permissions. To amend this, I used the `chmod u-w, g-w, g+r` command. This removed the write privileges for both the user and group and added the read permission for the group class. I then used `ls -la` to list the files and their permissions within the directory, including hidden files like .project_x.txt.

```
researcher2@ae520623ff19:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Oct  9 09:24 .
drwxr-xr-x 3 researcher2 research_team 4096 Oct  9 09:34 ..
-rw--w---- 1 researcher2 research_team  46 Oct  9 09:24 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Oct  9 09:24 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Oct  9 09:24 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Oct  9 09:24 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Oct  9 09:24 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Oct  9 09:24 project_t.txt
researcher2@ae520623ff19:~/projects$ chmod u-w,g-w,g+r .project_x.txt
researcher2@ae520623ff19:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Oct  9 09:24 .
drwxr-xr-x 3 researcher2 research_team 4096 Oct  9 09:34 ..
-r--r----- 1 researcher2 research_team  46 Oct  9 09:24 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Oct  9 09:24 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Oct  9 09:24 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Oct  9 09:24 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Oct  9 09:24 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Oct  9 09:24 project_t.txt
```

Change directory permissions

The researcher2 user should be the only one with permissions to access the drafts directory. As seen when first using the `ls -l` command to see the permissions, the group class has execute permissions to access the directory. This is changed when I used the `chmod g-x drafts` command to remove the execute permissions from the group class and the result is shown when using `ls -l` for a second time to confirm that the execute permission had been removed.

```
researcher2@ae520623ff19:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Oct  9 09:24 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Oct  9 09:24 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Oct  9 09:24 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Oct  9 09:24 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Oct  9 09:24 project_t.txt
researcher2@ae520623ff19:~/projects$ chmod g-x drafts
researcher2@ae520623ff19:~/projects$ ls -l
total 20
drwx----- 2 researcher2 research_team 4096 Oct  9 09:24 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Oct  9 09:24 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Oct  9 09:24 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Oct  9 09:24 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Oct  9 09:24 project_t.txt
researcher2@ae520623ff19:~/projects$
```

Summary

This project helped provide oversight into how to read permission strings, how to change permissions on files and directories, even if they are hidden and how to manage users, groups and others permissions from outside of a directory.